7. **Gallium is considered as ................ when added to the semiconductor.**

a. Acceptor atom  b. Donor atom  c. Both acceptor and donor.  d. Can not be determined.

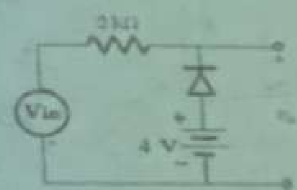
8. **For the following circuit, the status of the diode is ............................ (Assume a practical (simplified) model diode)**

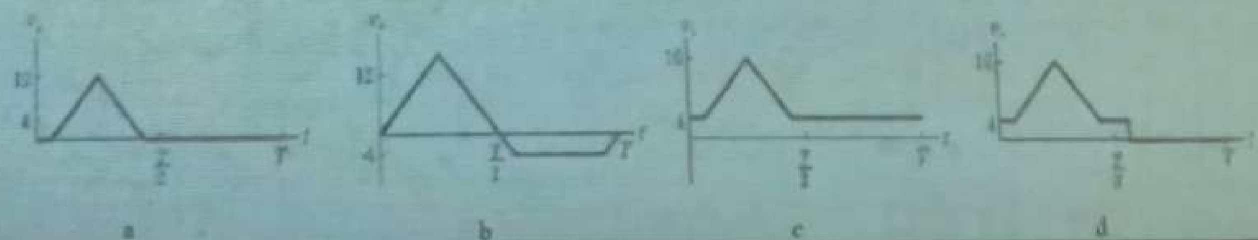
a. Forward-biased  b. Reverse-bias  c. Breakdown region  d. Not mentioned


9. **In a bridge configuration of full-wave rectifier and practical (simplified) model of Germanium (Ge) is used, the PIV for input voltage $V_{in}$ is given as .......**

a. Vin + 0.6 V  b. Vm − 0.7 V  c. Vin - 0.6 V  d. Vin + 0.7V  e. Vin − 0.3 V  f. Vin + 0.3 V

10. **For the given circuit below with ideal diode is used, the expected output for the given input $V_i$ is ....**

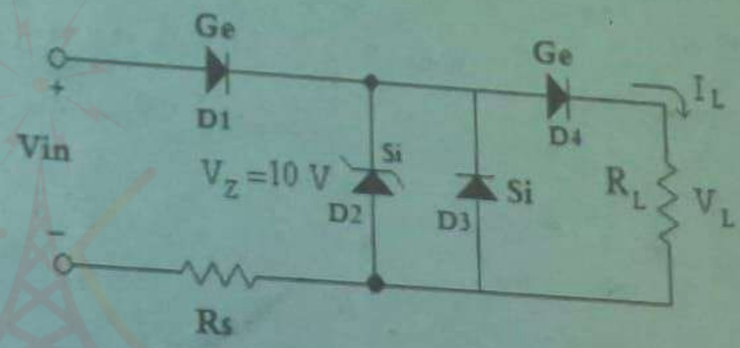
Q2. For the given circuit below, answer the following. Assume all the diodes are practical (simplified) model and the Zener diode is a Silicon diode. (4 Points)

Q2.1. For Vin = 8 V, find the value of $V_L$ for $R_s = 0\ \Omega$.

***The main equation used to find $V_L$:

...... $-8\text{ V} + 0.3\text{ V} + 0.3\text{ V} + V_L = 0$ ............

or $8\text{ V} - 0.3 - 0.3 - V_L = 0$




***The value of $V_L$: <u>Write the final answer only below</u>

......$V_L = 7.4$ V..................

**Q2.2. For Vin = 15 V, determine the value of the current $I_L$ for $R_L$ = 4.85 KΩ and $R_S$ = 100 Ω**

*****The main equation used to find $I_L$ (Other than the Ohm's law equation):**

............ $-10\ V + 0.3\ V + V_L = 0$ >> .........and $I_L = V_L / R_L$ ..........................................

Or $10 - 0.3 - V_L = 0$

*****The value of $I_L$: <u>Write the final answer only below</u>**

...............$I_L = 9.7\ V / 4.85\ K\Omega = 2\ mA$.........